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Import libraries and load the dataset

```python
import numpy as np
import pandas as pd
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import SimpleImputer, IterativeImputer

df = pd.read_csv(r"C:\Users\lexmi\OneDrive\Desktop\Auto_Missing.csv")

print(df.head())
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0          8         307.0       130.0  3504.0          12.0    70
1  15.0          8         350.0       165.0  3693.0          11.5    70
2  18.0          8         318.0       150.0  3436.0          11.0    70
3  16.0          8         304.0       150.0  3433.0          12.0    70
4  17.0          8         302.0       140.0  3449.0          10.5    70

   origin                       name
0     1.0  chevrolet chevelle malibu
1     1.0          buick skylark 320
2     1.0         plymouth satellite
3     1.0              amc rebel sst
4     1.0                ford torino
```

Mean Imputation (Horsepower)

```python
import numpy as np
import pandas as pd
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import SimpleImputer, IterativeImputer

df = pd.read_csv(r"C:\Users\lexmi\OneDrive\Desktop\Auto_Missing.csv")
mean_imputer = SimpleImputer(strategy="mean")
df["horsepower_Mean"] = mean_imputer.fit_transform(df[["horsepower"]])

print(df.head())
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0          8         307.0       130.0  3504.0          12.0    70
1  15.0          8         350.0       165.0  3693.0          11.5    70
2  18.0          8         318.0       150.0  3436.0          11.0    70
3  16.0          8         304.0       150.0  3433.0          12.0    70
4  17.0          8         302.0       140.0  3449.0          10.5    70

   origin                       name  horsepower_Mean
0     1.0  chevrolet chevelle malibu            130.0
1     1.0          buick skylark 320            165.0
2     1.0         plymouth satellite            150.0
3     1.0              amc rebel sst            150.0
4     1.0                ford torino            140.0
```

: Median Imputation (Weight)

```python
median_imputer = SimpleImputer(strategy="median")
df["weight_Median"] = median_imputer.fit_transform(df[["weight"]])
print(df.head())
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0          8         307.0       130.0  3504.0          12.0    70
1  15.0          8         350.0       165.0  3693.0          11.5    70
2  18.0          8         318.0       150.0  3436.0          11.0    70
3  16.0          8         304.0       150.0  3433.0          12.0    70
4  17.0          8         302.0       140.0  3449.0          10.5    70

   origin                       name  horsepower_Mean  weight_Median
0     1.0  chevrolet chevelle malibu            130.0         3504.0
1     1.0          buick skylark 320            165.0         3693.0
2     1.0         plymouth satellite            150.0         3436.0
3     1.0              amc rebel sst            150.0         3433.0
4     1.0                ford torino            140.0         3449.0
```

Mode Imputation (Origin)

```python
mode_imputer = SimpleImputer(strategy="most_frequent")
df["origin_Mode"] = mode_imputer.fit_transform(df[["origin"]]).ravel()
print(df.head())
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0          8         307.0       130.0  3504.0          12.0    70
1  15.0          8         350.0       165.0  3693.0          11.5    70
2  18.0          8         318.0       150.0  3436.0          11.0    70
3  16.0          8         304.0       150.0  3433.0          12.0    70
4  17.0          8         302.0       140.0  3449.0          10.5    70

   origin                       name  horsepower_Mean  weight_Median  \
0     1.0  chevrolet chevelle malibu            130.0         3504.0
1     1.0          buick skylark 320            165.0         3693.0
2     1.0         plymouth satellite            150.0         3436.0
3     1.0              amc rebel sst            150.0         3433.0
4     1.0                ford torino            140.0         3449.0

   origin_Mode
0          1.0
1          1.0
2          1.0
3          1.0
4          1.0
```

MICE Imputation

```python
num_cols = ["horsepower", "weight"]

mice_imputer = IterativeImputer(max_iter=10, random_state=42)

df_mice = df[num_cols].copy()
df_mice[:] = mice_imputer.fit_transform(df_mice)

df["horsepower_MICE"] = df_mice["horsepower"]
df["weight_MICE"] = df_mice["weight"]
print(df.head())
```

0.0s

```
    mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0  18.0          8         307.0       130.0  3504.0          12.0    70
1  15.0          8         350.0       165.0  3693.0          11.5    70
2  18.0          8         318.0       150.0  3436.0          11.0    70
3  16.0          8         304.0       150.0  3433.0          12.0    70
4  17.0          8         302.0       140.0  3449.0          10.5    70

   origin                       name  horsepower_Mean  weight_Median  \
0     1.0  chevrolet chevelle malibu            130.0         3504.0
1     1.0          buick skylark 320            165.0         3693.0
2     1.0         plymouth satellite            150.0         3436.0
3     1.0              amc rebel sst            150.0         3433.0
4     1.0                ford torino            140.0         3449.0

   origin_Mode  horsepower_MICE  weight_MICE
0          1.0            130.0       3504.0
1          1.0            165.0       3693.0
2          1.0            150.0       3436.0
3          1.0            150.0       3433.0
4          1.0            140.0       3449.0
```

Display the final DataFrame

```python
print(df.head(15))
```

[10] 0.0s

```
     mpg  cylinders  displacement  horsepower  weight  acceleration  year  \
0   18.0          8         307.0       130.0  3504.0          12.0    70
1   15.0          8         350.0       165.0  3693.0          11.5    70
2   18.0          8         318.0       150.0  3436.0          11.0    70
3   16.0          8         304.0       150.0  3433.0          12.0    70
4   17.0          8         302.0       140.0  3449.0          10.5    70
5   15.0          8         429.0       198.0  4341.0          10.0    70
6   14.0          8         454.0       220.0  4354.0           9.0    70
7   14.0          8         440.0       215.0  4312.0           8.5    70
8   14.0          8         455.0       225.0  4425.0          10.0    70
9   15.0          8         390.0       190.0  3850.0           8.5    70
10  15.0          8         383.0         NaN  3563.0          10.0    70
11  14.0          8         340.0       160.0  3609.0           8.0    70
12  15.0          8         400.0       150.0  3761.0           9.5    70
13  14.0          8         455.0       225.0  3086.0          10.0    70
14  24.0          4         113.0        95.0  2372.0          15.0    70

    origin                       name  horsepower_Mean  weight_Median  \
0      1.0  chevrolet chevelle malibu       130.000000         3504.0
1      1.0          buick skylark 320       165.000000         3693.0
2      1.0         plymouth satellite       150.000000         3436.0
3      1.0              amc rebel sst       150.000000         3433.0
4      1.0                ford torino       140.000000         3449.0
5      NaN           ford galaxie 500       198.000000         4341.0
6      1.0           chevrolet impala       220.000000         4354.0
...
11         1.0        160.00000       3609.0
12         1.0        150.00000       3761.0
13         1.0        225.00000       3086.0
14         3.0         95.00000       2372.0
```